

6.7 a

From 5.88

$$\left(\frac{1}{H_0} \frac{da}{dt}\right)^2 = \frac{\Omega_0}{a} + (1 - \Omega_0)$$

$$\left(\frac{da}{dt}\right)^2 = H_0^2 \left(\frac{\Omega_0}{a} + (1 - \Omega_0) \right)$$

$$\frac{da}{dt} = \sqrt{H_0^2 \left(\frac{\Omega_0}{a} + 1 - \Omega_0 \right)}$$

$$\frac{da}{dt} = H_0 \sqrt{\frac{\Omega_0}{a} + (1 - \Omega_0)}$$

From 5.89

$$H_0 t = \int_0^a \frac{1}{\sqrt{\Omega_0/a + 1 - \Omega_0}} da$$

$$\frac{d H_0 t}{da} = \frac{1}{\sqrt{\frac{\Omega_0}{a} + 1 - \Omega_0}}$$

$$H_0 \times \left(H_0 \frac{dt}{da} \right) = \left(\frac{1}{\sqrt{\frac{\Omega_0}{a} + 1 - \Omega_0}} \right)^{-1} \rightarrow \frac{da}{dt} = H_0 \sqrt{\frac{\Omega_0}{a} + (1 - \Omega_0)}$$

ii)

$$a(\theta) = \frac{1}{2} \frac{\Omega_0}{\Omega_0 - 1} (1 - \cos \theta) \quad 5.90$$

$$t(\theta) = \frac{1}{2H_0} \frac{\Omega_0}{(\Omega_0 - 1)^{3/2}} (\theta - \sin \theta) \quad 5.91$$

$$\frac{da}{dt} = \frac{da}{d\theta} \frac{d\theta}{dt} = \frac{da}{d\theta} \left(\frac{dt}{d\theta} \right)^{-1}$$

$$\frac{da}{d\theta} = \frac{\Omega_0}{2(\Omega_0 - 1)} \sin \theta$$

$$\frac{dt}{d\theta} = \frac{1}{2H_0} \frac{\Omega_0}{(\Omega_0 - 1)^{3/2}} (1 - \cos \theta)$$

$$\begin{aligned} \frac{da}{dt} &= \frac{\cancel{\Omega_0}}{2(\Omega_0 - 1)} \sin \theta \left(\frac{1}{2H_0} \frac{\cancel{\Omega_0}}{(\Omega_0 - 1)^{3/2}} (1 - \cos \theta) \right)^{-1} \\ &= H_0 \sqrt{\Omega_0 - 1} \frac{\sin \theta}{1 - \cos \theta} \end{aligned}$$

we know

$$\begin{aligned} a(\theta) &= \frac{1}{2} \frac{\Omega_0}{\Omega_0 - 1} (1 - \cos \theta) \rightarrow \frac{\Omega_0}{a} + 1 - \Omega_0 \\ &= \frac{2(\Omega_0 - 1)}{1 - \cos \theta} + 1 - \Omega_0 \end{aligned}$$

$$H_0 \sqrt{\frac{R_0}{a} + 1 - R_0} = H_0 \sqrt{R_0 - 1 \left(\frac{\sin^2 \theta}{1 - R_0} \right)}$$

6.8 b)

5.96

$$\frac{H^2}{H_0^2} = \frac{\Omega_{m,0}}{a^3} + 1 - \Omega_{m,0}$$

$$\left(\frac{H}{H_0}\right)^{-1} = \left(\sqrt{\frac{\Omega_{m,0}}{a^3} + 1 - \Omega_{m,0}}\right)^{-1} \quad \times \frac{1}{H_0}$$

$$a \frac{dt}{da} \approx \frac{1}{H} = \frac{1}{H_0} \sqrt{\frac{\Omega_{m,0}}{a^3} + 1 - \Omega_{m,0}} \quad \times H_0 da$$

$$H_0 dt = \sqrt{\frac{\Omega_{m,0}}{a^3} + 1 - \Omega_{m,0}} da$$

$$H_0 dt = \frac{da}{\sqrt{\frac{\Omega_{m,0} a^2}{a^3} + (1 - \Omega_{m,0}) a^2}}$$

$$= \frac{da}{\sqrt{\frac{\Omega_{m,0}}{a} + (1 - \Omega_{m,0}) a^2}}$$

$$H_0 t = \int_0^a \frac{da}{\sqrt{\frac{\Omega_{m,0}}{a} + (1 - \Omega_{m,0}) a^2}}$$

6.8b
(i)

$$a_m = \sqrt{\left(\frac{\Omega_{m0}}{1 - \Omega_{m0}} \right)}$$

5.100

we define

$$\frac{\Omega_{m0}}{a_m^3} = 1 - \Omega_{m0} \rightarrow \frac{\Omega_{m0}}{a} = (1 - \Omega_{m0}) \left(\frac{a}{a_m} \right)^3$$

$$H_0 t = \int_0^a \frac{da}{\sqrt{\frac{\Omega_{m0}}{a} + (1 - \Omega_{m0}) a^2}}$$

$$= \frac{1}{\sqrt{1 - \Omega_{m0}}} \int_0^a \frac{da}{\sqrt{\frac{\Omega_{m0}}{a(1 - \Omega_{m0})} + a^2}}$$

$$\frac{\Omega_{m0}}{1 - \Omega_{m0}} = a_m^3$$

$$= \frac{1}{\sqrt{1 - \Omega_{m0}}} \int_0^a \frac{da}{\sqrt{\frac{a_m^3}{a} + a^2}}$$

$$= \frac{1}{\sqrt{1 - \Omega_{m0}}} \int_0^a \frac{\sqrt{a} da}{\sqrt{a_m^3 + a^3}} = \frac{1}{\sqrt{1 - \Omega_{m0}}} \int_0^a \frac{da}{a_m^{3/2} a^{1/2} \sqrt{1 + \left(\frac{a}{a_m} \right)^3}}$$

iii)

$$H_0 t = \frac{1}{\sqrt{1-\Omega_{m_0}}} \int_0^a \frac{da}{a^{3/2-1/2} \sqrt{1+\left(\frac{a}{a_m}\right)^3}}$$

$$v = \sqrt{1+\left(\frac{a}{a_m}\right)^3}, \quad v^2 = 1+\left(\frac{a}{a_m}\right)^3$$

$$a = a_m \sqrt[3]{(v^2-1)}$$

$$\begin{aligned} da &= \frac{d}{dv} \left[a_m \sqrt[3]{(v^2-1)} \right] = a_m \frac{1}{3} (v^2-1)^{-2/3} \cdot 2v dv \\ &= \frac{2}{3} a_m v (v^2-1)^{-2/3} dv \quad (\text{into 1st eq}) \end{aligned}$$

$$H_0 t = \frac{1}{\sqrt{1-\Omega_{m_0}}} \int_0^a \frac{2 a_m v (v^2-1)^{-2/3} dv}{a^{3/2-1/2}}$$

$$= \frac{1}{\sqrt{1-\Omega_{m_0}}} \int_{v(0)=1}^v \frac{2}{3} \sqrt{\frac{a}{a_m}} (v^2-1)^{-2/3} dv$$

$$= \frac{1}{\sqrt{1-\Omega_{m_0}}} \int_{v(0)=1}^v \frac{2}{3} \sqrt{\frac{a_m \sqrt[3]{(v^2-1)}}{a_m}} (v^2-1)^{-2/3} dv$$

$$= \frac{1}{\sqrt{1-\Omega_{m_0}}} \int_{v(0)=1}^v \frac{2}{3} (v^2-1)^{1/6} (v^2-1)^{-2/3} dv$$

$$= \frac{2}{3 \sqrt{1-\Omega_{m_0}}} \int_{v(0)=1}^v \frac{dv}{\sqrt{v^2-1}}$$

$$\text{iv)} \int \frac{du}{\sqrt{u^2 - 1}}$$

$$u = \sec \theta = \sqrt{1 + \left(\frac{a}{am}\right)^2}$$

$$du = \sec \theta \tan \theta d\theta$$

$$\int \frac{\sqrt{1 + \left(\frac{a}{am}\right)^2} du}{\sqrt{\sec^2 \theta - 1}} = \int \frac{\sqrt{1 + \left(\frac{a}{am}\right)^2} du}{\tan \theta}$$

$$= \int \frac{\sqrt{1 + \left(\frac{a}{am}\right)^2} \sec \theta \tan \theta d\theta}{\tan \theta}$$

$$= \ln \left| \sec \theta + \tan \theta \right| + C$$

$$\tan \theta = \sqrt{\sec^2 \theta - 1} = \sqrt{u^2 - 1} = \sqrt{\left(\frac{a}{am}\right)^2}$$

$$= \ln \left[\sqrt{1 + \left(\frac{a}{am}\right)^2} + \left(\frac{a}{am}\right)^{1/2} \right] - \ln \left(\frac{u(a) + \sqrt{u^2 - 1}}{u(a) + \sqrt{u_0^2 - 1}} \right)$$

$= \ln(1 + \sqrt{1+1})$
 $= \ln(1) = 0$

$$H_0 t = \frac{2}{3\sqrt{1 - \Omega_{m_0}}} \ln \left[\sqrt{1 + \left(\frac{a}{am}\right)^2} + \left(\frac{a}{am}\right)^{1/2} \right]$$

V

$$\ln(1+x) \approx x \quad \text{For } x \ll 1$$

if

$$a \ll a_m$$

then

$$\left(\frac{a}{a_m}\right)^3 \ll 1$$

$$\sqrt{1 + \left(\frac{a}{a_m}\right)^3} \approx 1 + \frac{1}{2} \left(\frac{a}{a_m}\right)^3$$

$$H_0 t \approx \frac{2}{3\sqrt{1-\Omega_{m_0}}} \ln \left[\left(\frac{a}{a_m}\right)^{3/2} + 1 + \frac{1}{2} \left(\frac{a}{a_m}\right)^3 \right]$$

$$x = \left(\frac{a}{a_m}\right)^{3/2} \rightarrow H_0 t \approx \frac{2}{3\sqrt{1-\Omega_{m_0}}} \left(\frac{a}{a_m}\right)^{3/2}$$

$$\left(\frac{a}{a_m}\right)^{3/2} \approx \frac{3}{2} \sqrt{1-\Omega_{m_0}} H_0 t$$

$$\rightarrow a(t) \approx \left(\frac{3}{2} \sqrt{1-\Omega_{m_0}} \right)^{2/3}$$

$$\rightarrow a_m = \left(\frac{\Omega_{m_0}}{1-\Omega_{m_0}} \right)^{1/3}$$

$$a(t) = \sqrt[3]{\frac{\Omega_{m_0}}{1-\Omega_{m_0}}} \left(\frac{3}{2} \sqrt{1-\Omega_{m_0}} H_0 t \right)^{2/3} = \left(\frac{3}{2} \sqrt{\Omega_{m_0}} H_0 t \right)^{2/3}$$

Vi

$$H_0 t = \frac{2}{3\sqrt{1-\Omega_{m_0}}} \ln \left[\left(\frac{a}{a_m} \right)^{3/2} + \sqrt{1 + \left(\frac{a}{a_m} \right)^3} \right]$$

$$a_m = \left(\frac{\Omega_{m_0}}{1-\Omega_{m_0}} \right)^{1/3}$$

$$\frac{1}{a_m^3} = \frac{1-\Omega_{m_0}}{\Omega_{m_0}}$$

$$\left(\frac{1}{a_m} \right)^{3/2} = \left(\frac{1-\Omega_{m_0}}{\Omega_{m_0}} \right)^{1/2}$$

$$H_0 t = \frac{2}{3\sqrt{1-\Omega_{m_0}}} \ln \left[\left(\frac{1}{a_m} \right)^{3/2} + \sqrt{1 + \frac{1}{a_m^3}} \right]$$

$$H_0 t = \frac{2}{3\sqrt{1-\Omega_{m_0}}} \ln \left[\left(\frac{1-\Omega_{m_0}}{\Omega_{m_0}} \right)^{1/2} + \sqrt{1 + \frac{1-\Omega_{m_0}}{\Omega_{m_0}}} \right]$$

$$H_0 t = \frac{2}{3\sqrt{1-\Omega_{m_0}}} \ln \left(\frac{\sqrt{1-\Omega_{m_0}}}{\sqrt{\Omega_{m_0}}} + \frac{1}{\sqrt{\Omega_{m_0}}} \right)$$

$$H_0 t = \frac{2}{3\sqrt{1-\Omega_{m_0}}} \ln \left(\frac{\sqrt{1-\Omega_{m_0}} + 1}{\sqrt{\Omega_{m_0}}} \right)$$

$$H_0 t = \frac{2}{3\sqrt{1-\Omega_{m_0}}} \ln \left(\frac{\sqrt{1-\Omega_{m_0}} + 1}{\sqrt{\Omega_{m_0}}} \right)$$

$$\Omega_{m_0} = 0.32$$

$$\frac{1}{H_0} = 14.6 \text{ Gyr}$$

$$t_0 = \frac{2 (14.6 \times 10^9)}{3\sqrt{1-0.32}} \ln \left(\frac{\sqrt{1-0.32} + 1}{\sqrt{0.32}} \right)$$

$$= 13.82 \text{ Gyr}$$

Viii)

matter and Λ are equal when $a = a_{m\Lambda} = a$

$$\frac{\Omega_{m_0}}{a^3} = \Omega_{\Lambda_0} = 1 - \Omega_{m_0}$$

$$a_{m\Lambda} = \sqrt[3]{\left(\frac{\Omega_{m_0}}{1 - \Omega_{m_0}}\right)} = a_m$$

$$H_0 t_1 = \frac{2}{3\sqrt{1 - \Omega_{m_0}}} \ln \left[1^{3/2} + \sqrt{1 + 1^3} \right]$$
$$= \frac{2}{3\sqrt{1 - \Omega_{m_0}}} \ln(1 + \sqrt{2})$$

$$t_1 = \frac{2 H_0^{-1}}{3\sqrt{1 - \Omega_{m_0}}} \ln(1 + \sqrt{2})$$

$$H_0^{-1} = 14.6 \text{ Gyr}$$

$$\Omega_{m_0} = 0.32$$

$$t_1 = \frac{2 \times 14.6}{3\sqrt{1 - 0.32}} \ln(1 + \sqrt{2}) = 10.403$$

ix

$$a = 1 \quad (\text{present time})$$

$$a(t) = f(H_0(t-t_0))$$

Flat matter $\Lambda > 0$ universe

$$\Omega_{m,0} = 0.32$$

should be concave up, expanding faster with time as Λ dominates over matter

6.9

$$i) \quad \frac{H^2}{H_0^2} = \frac{\Omega_{r,0}}{a^4} + \frac{\Omega_{m,0}}{a^3}$$

$$a_{rm} = \frac{\Omega_{r,0}}{\Omega_{m,0}}$$

$$\frac{\Omega_{m,0}}{a^3} = \frac{\Omega_{r,0}}{a_{rm} a^3}$$

$$\frac{H^2}{H_0^2} = \frac{\Omega_{r,0}}{a^4} + \frac{\Omega_{r,0}}{a_{rm} a^3} = \frac{\Omega_{r,0}}{a^4} \left(1 + \frac{a}{a_{rm}} \right)$$

$$H = H_0 \sqrt{\frac{\Omega_{r,0}}{a^4} \left(1 + \frac{a}{a_{rm}} \right)}$$

$$H = \frac{da}{a dt} = H_0 \sqrt{\frac{\Omega_{r,0}}{a^4} \left(1 + \frac{a}{a_{rm}} \right)}$$

$$dt = \frac{a da}{H_0 \sqrt{\frac{\Omega_{r,0}}{a^4} \left(1 + \frac{a}{a_{rm}} \right)}}$$

$$dt = \frac{a da}{\Omega_{r,0}^{1/2}} \left[1 + \frac{a}{a_{rm}} \right]^{-1/2}$$

(i)

$$H_0 t = \frac{4a^2 r_m}{3\sqrt{\Omega_{r0}}} \left[1 - \left(1 - \frac{a}{2a_{rm}} \right) \left(1 + \frac{a}{a_{rm}} \right)^{1/2} \right]$$

$$\frac{da}{dt} = \Omega_{r0}^{1/2} \sqrt{\left(1 + \frac{a}{a_{rm}} \right)}$$

$$H_0 dt = \frac{a da}{\sqrt{\Omega_{r0}}} \sqrt{1 + \frac{a}{a_{rm}}}$$

$$f(a) = \left(1 - \frac{a}{2a_{rm}} \right) \sqrt{1 + \frac{a}{a_{rm}}}$$

$$\frac{df}{da} = -\frac{1}{2a_{rm}} \sqrt{1 + \frac{a}{a_{rm}}}$$

$$\frac{d}{da} (H_0 t) = \frac{4a^2 r_m}{3\sqrt{\Omega_{r0}}} \left(-\frac{df}{da} \right)$$

$$\frac{dt}{da} = \frac{a}{\sqrt{\Omega_{r0}}} \left(1 + \frac{a}{a_{rm}} \right)^{1/2}$$

$$H_0 dt = \frac{a da}{\sqrt{\Omega_{r0}}} \left(1 + \frac{a}{a_{rm}} \right)^{-1/2}$$

iii)

$$H_{ot} = \frac{u a_{rm}^2}{3\sqrt{5}r_0} \left[1 - \left(1 - \frac{a}{2a_{rm}}\right) \left(1 + \frac{a}{a_{rm}}\right)^{1/2} \right]$$

Taylor Series:

$$(1+x)^{1/2} \approx 1 + \frac{x}{2} - \frac{x^2}{8}$$

$$x = \frac{a}{a_{rm}}$$

$$\rightarrow 1 - \left(1 - \frac{a}{2a_{rm}}\right) \left(1 + \frac{a}{2a_{rm}} - \frac{a^2}{8a_{rm}^2}\right)$$

(to 2nd order approx.)

$$= 1 - \left(1 - \frac{a^2}{8a_{rm}^2} - \frac{a^2}{4a_{rm}^2} + \cancel{\frac{a}{2a_{rm}}} - \cancel{\frac{a}{2a_{rm}}}\right)$$

$$= \frac{3}{8} \frac{a^2}{a_{rm}^2}$$

$$H_{ot} = \frac{4 \cancel{a_{rm}^2}}{3\sqrt{5}r_0} \left(\frac{3}{8} \frac{a^2}{\cancel{a_{rm}^2}} \right) = \frac{a^2}{2\sqrt{5}r_0}$$

$$a(t) = \sqrt{2H_{ot}\sqrt{5}r_0}$$

iv)

$$H_{\text{tot}} = \frac{U a r_m^2}{3 \sqrt{\Omega r_0}} \left[1 - \left(1 - \frac{a}{2 a r_m} \right) \left(1 + \frac{a}{a r_m} \right)^{1/2} \right]$$

$$a \gg a r_m, \quad \frac{a}{a r_m} \gg 1$$

$$\left(1 + \frac{a}{a r_m} \right) \approx \frac{a}{a r_m}$$

$$H_{\text{tot}} = \frac{U a r_m^2}{3 \sqrt{\Omega r_0}} \left[1 - \left(\frac{a}{2 a r_m} \right) \left(\frac{a}{a r_m} \right)^{1/2} \right]$$

$$= \frac{U a r_m^2}{3 \sqrt{\Omega r_0}} \left[1 - \frac{1}{2} \left(\frac{a}{a r_m} \right)^{3/2} \right] \quad \left(\frac{a}{a r_m} \right)^{3/2} \gg 1$$

$$= \frac{2}{3} \frac{a^{1/2}}{\sqrt{\Omega r_0 a r_m}}$$

$$a(t) = \frac{3}{2} H_{\text{tot}} \sqrt{\Omega r_0 a r_m}$$

V

$$H_0 t = \frac{u a_{rm}^2}{3\sqrt{5}c r_0} \left[1 - \left(1 - \frac{a}{2a_{rm}}\right) \left(1 + \frac{a}{a_{rm}}\right)^{1/2} \right]$$

$$a = a_{rm}$$

$$H_0 t = \frac{u a_{rm}^2}{3\sqrt{5}c r_0} \left[1 - \left(1 - \frac{1}{2}\right)^{1/2} \left(1 + 1\right)^{1/2} \right]$$

$$= \frac{u a_{rm}^2}{3\sqrt{5}c r_0} \left[1 - \frac{\sqrt{2}}{2} \right]$$

$$t = \frac{u a_{rm}^2}{3\sqrt{5}c r_0 H_0} \left[1 - \frac{\sqrt{2}}{2} \right]$$

Vi

$$\Omega_{r_0} = 9 \times 10^{-5}$$

$$\Omega_{m_0} = 0.32$$

$$H_0^{-1} = 14.6 \text{ Gyr}$$

$$t = \frac{4 a_{rm}^2}{3 \sqrt{\Omega_{r_0}} H_0} \left[1 - \frac{\sqrt{2}}{2} \right]$$

by definition

$$a_{rm} = \frac{\Omega_{r_0}}{\Omega_{m_0}}$$

$$t = \frac{4 \Omega_{r_0}^{3/2}}{3 \sqrt{\Omega_{r_0}} H_0 \Omega_{m_0}^2} \left[1 - \frac{\sqrt{2}}{2} \right]$$

$$= 4.75 \times 10^5 \text{ Gyr}$$

vii)

by def.

$$z_{rm} = \frac{1}{a_m} - 1 = \frac{1}{\frac{\Omega_{r_0}}{\Omega_m}} - 1$$

$$= \frac{\Omega_m}{\Omega_{r_0}} - 1 = \frac{0.32}{9 \times 10^5} - 1 = 3554.6$$